GlucoTwin: system architecture

Solid green = real data · dashed orange = simulated

Digital twin that fuses static EHR data with dynamic CGM / wearable streams to forecast a glucose spike (>180 mg/dL) 2 hours ahead in type 2 diabetes

1 · DATA SOURCES

Dynamic stream: CGM + logs
Shanghai T2DM dataset, 100 patients, 109 recordings. Glucose every 15 min, timed meals, insulin doses.
REAL

Static stream: clinical record
Demographics, diabetes duration, HbA1c, labs (C-peptide, eGFR, lipids), medications, complications.
REAL

Wearables
Heart rate and step count generated from time of day, age, BMI and logged meals. Never reads glucose (no leakage).
SIMULATED

2 · DATA PIPELINE (SRC/BUILD_DATASET.PY)

Clean & align
Snap to 15-min grid, interpolate gaps up to 30 min, split on longer gaps.

Parse events
Meal text to grams of food and carbohydrate; insulin text to IU; "has insulin record" flag.

Static features
HbA1c to %, medication and comorbidity flags, labs.

Sliding windows
2 h look-back, 2 h horizon. Only windows with glucose ≤180 now, so the task is "will it cross?". 89,974 windows, 14.8% positive.

3 · FUSION & MODEL (SRC/TRAIN.PY)

Feature fusion
18 dynamic features (trend, variability, time since meal / insulin, carbs, time of day) + static clinical features + simulated wearables.

LightGBM classifier
Outputs risk 0-1 that glucose exceeds 180 mg/dL within 120 min. MLP fusion network trained for comparison.

Evaluation
Patient-grouped 5-fold CV (no patient in both train and test), patient-level bootstrap CIs, ablation: glucose only / dynamic / static / fused / + wearables.

Alert logic (src/alert_eval.py)
Threshold + 2 h silence per alert; alerts/day, precision, spikes warned.

4 · DASHBOARD (APP/DASHBOARD.PY)

Virtual patient view
Patient card, glucose timeline with meals, insulin and alerts; risk curve; simulated HR and steps.

Forecast & alert
2-hour risk, High / Moderate / Low, replay of what really happened next.

Explainability
Top factors raising or lowering this patient's risk (tree contributions).

What-if & sensitivity
Add / remove a meal or insulin and see the model's response; doctor sets the alert threshold.

Results (held-out patients)
AUROC 0.73 glucose only → 0.86 with meals, insulin and trend. At threshold 0.5: ~2.5 alerts/day, 63% correct, 78% of spikes warned.

Honest limits
Shanghai cohort, 3-14 days per patient. Heart rate and steps are simulated. Clinical record adds no accuracy once dynamic data is present.

Stack & license
Python, pandas, scikit-learn, LightGBM, Plotly, Streamlit. MIT license. Research prototype, not a medical device.